

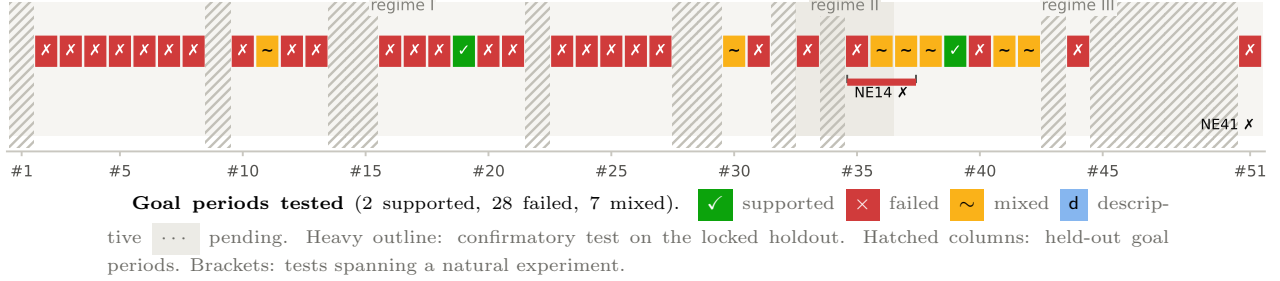
H42: Cross-excitation is a delayed step at the next read-out

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1/5 Usefulness 2/5

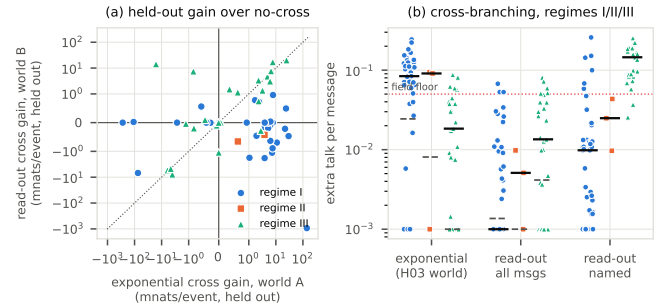
Bottom line. A read-out-anchored kernel reveals no hidden coupling: each agent’s own call clock sets talk timing, and real message-driven excitation is confined to messages that name the recipient.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H42 redoes H03’s multivariate Hawkes fits with kernels anchored at the recipient’s next model call, the read-out call from the context ledger. The fits cover 71 non-holdout period units in 35 periods, with synthetic validation, shift and day-block nulls, and four native tests. A read-out kernel does not raise cross-excitation; it removes most of it. Talk locks to each agent’s *own* call clock. A model in which talk can only follow the recipient’s call starts wins by 1.1 nats per held-out message in 57/57 units. Inside that model, messages read at a call barely change whether the agent talks: $n_{\text{cross}} \approx 0.004$, against 0.061 for H03’s exponential kernel. Regime I is ≈ 0 ; regime III is 0.01–0.05. The pre-registered estimator “confirms” the prediction but fails the synthetic guard. Natives NE41, NE14 and #51 fail. Post hoc, messages that *name* the recipient carry 0.15 extra talk events each in regime III. The holdout is not run.



Observables. (a) Per unit, the held-out gain from cross-excitation: the read-out kernel in the call-clock world against the exponential kernel in the H03 world (symlog). The read-out gain is ≤ 0 in most regime-I units. (b) Extra talk per message by regime (I, II, III): exponential (H03 world), read-out over all messages, and read-out for messages that name the recipient (post hoc). Bars: medians. Dashed: shift-null q95; 0.001 = zero. Red: synthetic shared-field floor.

PHYSICAL MODEL

Model 09 (multivariate Hawkes) in two worlds: λ^A is the H03 world and λ^B the call-clock world. Events are agent chat messages. Sources are other agents’ messages that agent i read (ledger pairs). c_k is the start of i ’s k -th call, and k^* the latest. f_i is the talk-delay density, and κ the call class (chat or computer use, wake or not). $k_1(m)$ is the call that read message m . w_b are read-out weights over call-index bins b ; α_q, β_q form an exponential grid. X_i is the cross part, and n_{cross} the cross offspring per event.

$$\lambda_i^A = c_{id} s_b + \Phi_i^{\text{self}} + \sum_m \sum_q \alpha_q \beta_q e^{-\beta_q(t - s_m)}$$

$$\lambda_i^B = f_i(t - c_{k^*}) \left[c_{id} \kappa s_b + \Phi_i^{\text{self}} + \sum_b \frac{w_b}{M_b} N_b(k^*) \right]$$

$$N_b(k^*) = \#\{m : k^* - k_1(m) \in b\}, \quad k_1(m) = \min\{k : c_k > s_m\}$$

$$n_{\text{cross}} = \frac{1}{N} \sum_i \int X_i(t) dt$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Before you fit an influence model to agent logs, model each agent’s call schedule and call class. Event-timing “contagion” between LLM agents mostly follows each agent’s own call clock. Without the call clock, branching ratios and influence graphs from timestamps mostly measure the scaffold, and shift nulls do not catch this. With it, an ordinary message barely changes whether an agent speaks. The one handle left is direct address. In the always-on regime, a message that names an agent adds about 0.15 talk events at its next call (post hoc; mid-exchange confound open). To get a response, name the agent, as H29 also finds. The read-out kernel itself is not a new knob.

H42-readout-hawkes-kernel · generated 2026-10-04 12:01 by infra/summaries/build_summaries.py · card hypotheses/H42-readout-hawkes-kernel/README.md · ratings: H42 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed (amended reading; Verdict pre-registered in brackets)	
P1 read-out beats exponential, held out	gain > exponential gain in 33% (III 52%) [pr 89%, void]	failed
P3 exponential understates n_{cross}	0.004 vs 0.061; ratio 0.22 [pr 0.41 vs 0, void]	reversed
P4 consistent across regimes	regime medians 0 / 0.005 / 0.013 (exp. 0.084 / 0.091 / 0.018)	failed
P5 step at calls 0–1	100% of mass at calls 0–1, 73% at read-out (34 units)	supported
P6 beats nulls	shift q95 42%, day-block mean 63%	failed
P8 call index beats time since arrival	$B \geq A_g$ in 40%	failed
G1 synthetic guard	shared field passes every null; pr inflates n_x 70%	failed
NE41 erasure cuts tail	surviving fraction 3.1 ± 2.5 (pred. ≤ 0.5)	failed
G19 regime-I talk ignores reads	$n_x = 0.000$; pr $0.89 \approx$ its shift null 0.88	supported
post hoc: named messages	0.15 per named msg (III), > null 25/27; unnamed ≈ 0	frozen (C5)

GOAL PERIODS IN DETAIL

G02 $\times$ failed (n_x read-out 0.022 vs exponential 0.166; held...
G03 $\times$ failed (n_x read-out 0.067 vs exponential 0.141; held...
G04 $\times$ failed (n_x read-out 0.030 vs exponential 0.121; held...
G05 $\times$ failed (n_x read-out 0.026 vs exponential 0.087; held...
G06 $\times$ failed (n_x read-out 0.000 vs exponential 0.129; held...
G07 $\times$ failed (n_x read-out 0.000 vs exponential 0.126; held...
G08 $\times$ failed (n_x read-out 0.000 vs exponential 0.016; held...
G10 $\times$ failed (n_x read-out 0.010 vs exponential 0.078; held...
G11 $\sim$ mixed (n_x read-out 0.013 vs exponential 0.071; held...
G12 $\times$ failed (n_x read-out 0.006 vs exponential 0.099; held...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

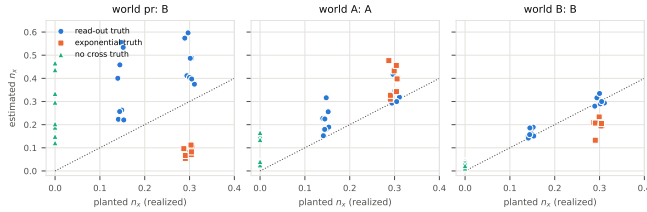
The call-clock world takes call times and call classes as given. It therefore excludes by construction any influence that changes *when* or *how* an agent calls; the activity channel is only a partial check. Outside Gemini, call starts are calibrated, not measured (regime-I chat calls are latency-placed); bound sensitivity changes nothing. Shift and day-block nulls do not control a shared 15-min field. The card has three dated amendments (two estimator worlds, the field floor); it reports the pre-registered verdicts and voids them. The named-message result is post hoc, and recipients mid-exchange are its main rival.

ROUND 2

Run `confirm.py` (C1–C5: #51 tail, #28, #22, #29a, #43). Separate named-message triggering from ongoing exchanges with DQ2 reply threads. Model call timing as its own process, to measure excitation through call timing or call class.

- **Essence**. Whether, and through which messages, what an agent reads changes what it does next, measured on the recipient's call clock.
- **H42-R1** Named messages as the coupling carrier.
- **H42-R2** Two-layer point process.
- **H42-R3** Shared-field control.

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Second figure. Synthetic validation on the real call grids of #27, #33, #40 and #51c, with a shared 15-min field in every arm (32 runs). Only the call-clock world recovers planted read-out excitation (median +6%). With no cross-excitation, every world still beats its empirical nulls (field floor 0.05).